

PA4 - HI-FI PROTOTYPE

Course: Interface Design (CSC13112)

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Group: Group 06 - **Class:** 23KTMP2

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Live Prototype: testpa4.vercel.app

Demo Video: www.youtube.com/watch?v=pGMSQEdcXys

1. Introduction & Design Goals

After the formative usability testing process with 6 paper-prototype variants in PA3, Group 06 selected and merged two design solutions:

1. **Nav-1 (Fixed Bottom Navigation Bar):** A fixed 5-tab navigation bar at the bottom of the screen, thoroughly addressing thumb-zone reachability.
2. **Sch-2 (Date Strip Filter + Dedicated 8×8 Match Detail):** An intuitive date-strip slider combined with compact match cards and a dedicated 8×8 chessboard-position analysis screen.

In PA4, the team developed this solution into a complete **High-Fidelity Prototype** that supports direct interaction with realistic data across mobile devices.

2. Technical Architecture & Implementation Technologies

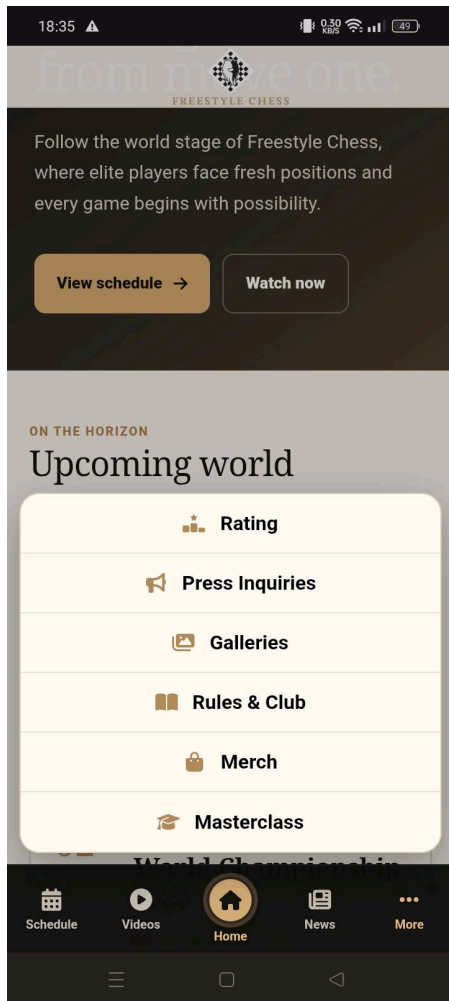
The prototype is not a static mockup; it is implemented as a real interactive web application:

- **Core Framework:** React 19 combined with TypeScript to ensure type safety and a modular component structure.
- **Build Tool & Runtime:** Vite for fast startup, instant hot reloading, and smooth 60fps rendering performance on mobile.
- **Routing:** **React Router** with **HashRouter** for zero-latency navigation and flexible static deployment.
- **Design Tokens & Styling:** Modern vanilla CSS (CSS Custom Properties), with CSS Grid/Flexbox layouts optimized for mobile screen standards.
- **Advanced interactive components:** Radix Popover and React Day Picker for the Date Picker Modal.
- **Near-realistic data:** Integrated tournament information such as the *2026 FIDE Freestyle Chess World Championship*, ELO ratings, portraits of international Grandmasters (Magnus Carlsen, Le Quang Liem, Hikaru Nakamura, Fabiano Caruana, Vincent Keymer, etc.), and real match chessboards from chess.com.

3. Main Components in the Prototype

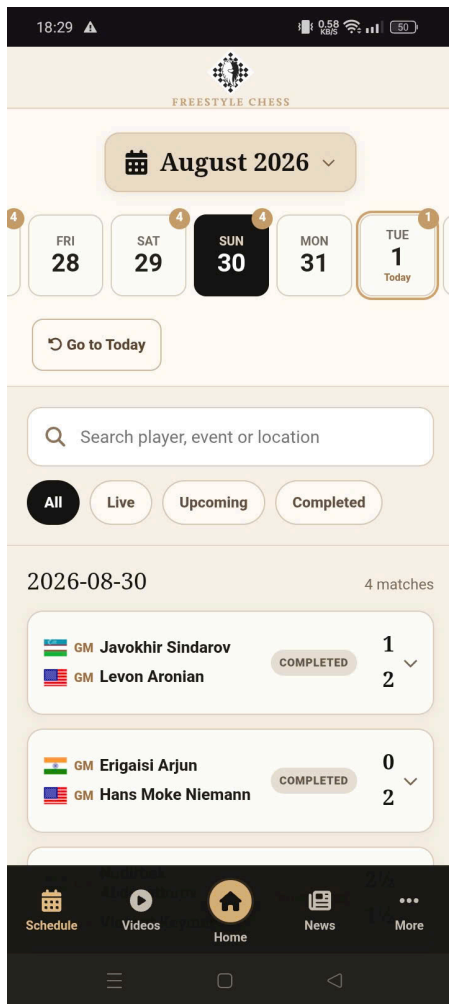
3.1. Navigation Bar

- **Fixed Bottom Navigation (Nav-1):** 5 fixed tabs (*Schedule, Videos, Home, News, More*) at the bottom edge of the screen, placed within the thumb-reach zone with standard touch target sizes of $\geq 48 \times 48$ px.
- **More Action Sheet:** Tapping the 'More' tab opens a menu for secondary pages (Rating, Galleries, Merch, etc.) without covering the main navigation bar. Users can tap anywhere on the corresponding button to open a secondary page.



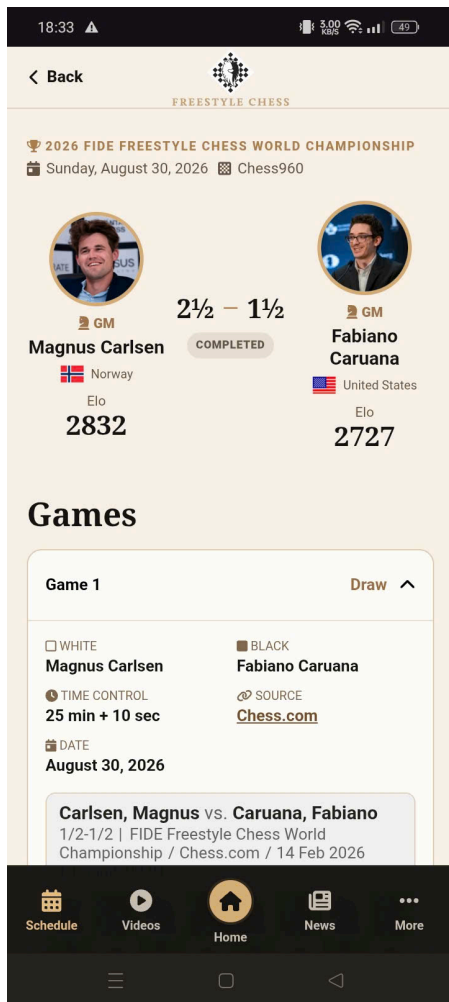
3.2. Schedule Screen

- **Top App Bar:** Page title, search button, and Date Picker button.
- **Horizontal Date Strip:** A horizontally scrollable date strip that allows users to select any day in the week. The selected date tab is highlighted.
- **Date Picker Modal:** Allows users to quickly jump to distant dates in the month, with a "Go to Today" button to return to the current date.
- **Instant Search & Filter Chips:** Instant search by player name/location combined with status filters (*Live, Upcoming, Completed*).
- **Expandable Matchup Cards:** Match cards display national flags, the names of the two players, ELO ratings, match status, and a "View Details →" button.



3.3. Match Detail Screen

- **Sticky Header & Contextual Back Button:** A clear Back button in the top-left corner helps users return to the exact date and filter state on the Schedule page without losing context.
- **Player Banner:** Displays information about the two players, ELO ratings, score, and match status.
- **Interactive 8×8 Chessboard:** A visual 8×8 chessboard that accurately shows the real position of each chess game.



4. Four Ergonomic & Accessibility Criteria

1. **Touch Target Size** $\geq 48 \times 48$ px: Completely eliminates thumb mis-tap errors.
2. **Thumb Zone Optimization**: Places the most frequently used functions in the lower half of the screen (Natural Thumb Zone).
3. **Contrast Ratio** $\geq 4.5 : 1$: Targets WCAG AA contrast for normal text; the clean Light Mode background supports readability under outdoor lighting.
4. **Instant Feedback** < 250 ms: All micro-interactions respond instantly within 250ms, creating a smooth Native App-like experience.